A design proposal of interferometers on silicon photonics

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***Index Terms*— Silicon Photonics, Mach-Zehnder Interferometer, Michelson Interferometer**

I. INTRODUCTION

Interferometers are of significant use on the optical bench. We use the interferometers to accurately measure small differences of optical paths. Scaling down from meters to micrometers, we can build such intricate systems using semiconductor manufacturing technologies. Besides being the proof-of-concept realization, we can use interferometers to design functional photonic chips.

The proposed design will be implemented with a silicon waveguide of height 220 nm and width 500 nm, and with TE polarization. The design includes three parts:

1. Mach-Zehnder interferometers (MZI) with varying length differences. This part is to examine if the predicted signal patterns match the experiment results.
2. Michelson interferometers with loop mirrors. This part is to examine if we can get the halved free spectral range (FSR) as predicted.
3. Simple in/out circuits with varying path lengths to characterize the insertion loss and waveguide performance. This part acts as testing loops.

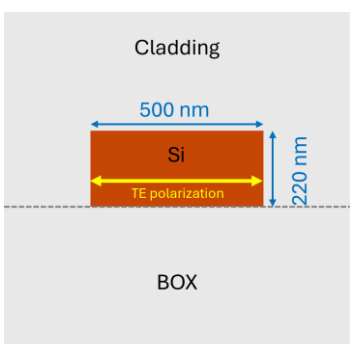

Fig. 1. Geometry of the waveguide (width 500 nm, height 220 nm) and the polarization direction.

II. THEORY

A. Waveguide

A photonic waveguide is composed of a higher index core material and a lower index cladding material. The dimensions and index difference determine the degree of confinement of the optical mode. Normally, the waveguide is designed to be rectangular. For a common SOI process, the height is 220 nm and the width is around 500 nm.

The propagation mode often has a bandwidth, and thus will experience a dispersion. The information is propagating at a velocity called the group velocity, labeled as n_g.

$$n_g = n - \lambda \frac{dn}{d\lambda} \tag{1}$$

To change the direction of light, we use bent waveguides. Bent structure will introduce bent loss and thus need careful modeling to optimize for the design.

B. Y-Branch splitter

To create different paths, we need splitters. The y-branch splitter has two directions: splitting and combining.

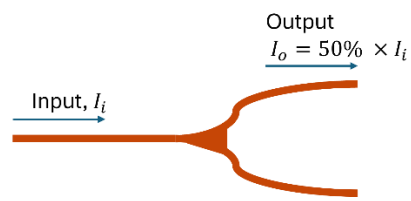

Fig. 2. A 50-50 y-branch splitter.

For an ideal 50-50 y-branch splitter, the output power from of one the port is 50% of the input power.

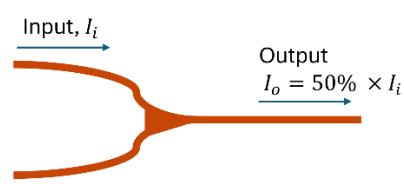

Fig. 3. A 50-50 y-branch combiner.

For a combiner, however, it's important to remember that the input from one port only transmits 50% of the power to the fundamental mode, another 50% to the higher order or radiation mode. The reciprocity of the 50-50 splitting in the case of the combiner thus be port-to-modes, instead of the port -to-ports for the splitter.

We can use the y-branch to split and combine light and build interferometers.

C. Interferometers

Interferometers are devices that combine multiple waves to create interference patterns, from which we can measure quantities like distance, optical index, etc.

Mach-Zehnder interferometers (MZI) constitutes of two beam splitters. The first is to create two lights from one input and the second then combine them into one output.

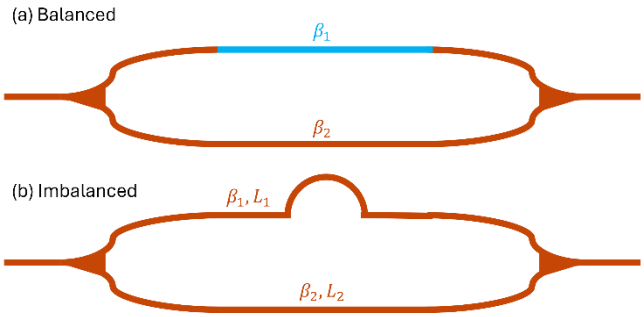

Fig. 4. MZI model. (a) Balanced MZI. (b) Imbalanced MZI.

It's balanced MZI if the two paths are the same, and imbalanced if they are different. The transfer function of MZI is

$$\frac{I_o}{I_i} = \frac{1}{4}\left|e^{-i\beta_1 L_1 - \frac{\alpha_1}{2}L_1} + e^{-i\beta_2 L_2 - \frac{\alpha_2}{2}L_2}\right|^2 \quad (2)$$

where $\beta_i = 2\pi n_i / \lambda$ is the propagation constant, n_i is the effective index, α_i is the loss, and L_i is the length of each waveguide.

For lossless waveguides, the transfer function can be written as

$$\frac{I_o}{I_i} = \frac{1}{2}[1 + cos(\beta_1 L_1 - \beta_2 L_2)] \quad (3)$$

If the two waveguide have the same propagation constant but differ in length, as the model shown in Fig 4 (b), then the transfer function is

$$\frac{I_o}{I_i} = \frac{1}{2}\left[1 + cos\left(\frac{2\pi n}{\lambda}\Delta L\right)\right] \quad (4)$$

where $\Delta L = L_1 - L_2$ is the length difference.

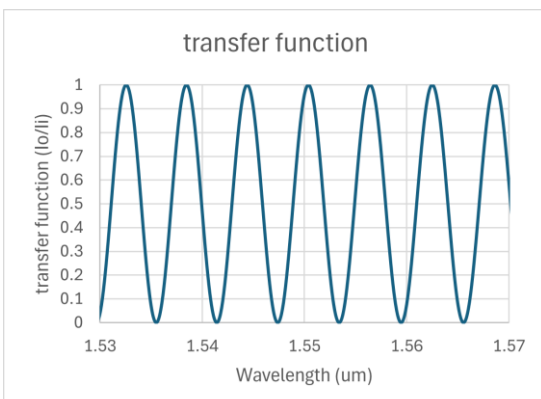

Fig. 5. Plot of the transfer function. Assume the index at 1.5 um is 2.4, with dn/dλ = -1 μm^{-1}, $\Delta L = 100\ \mu m$.

The transfer function changes sinusoidally with wavelength. The peak-to-peak spacing is call the free spectral range (FSR). Around a particular peak, we can find the FSR by

$$FSR = \frac{\lambda^2}{\Delta L n_g} \quad (5)$$

We can also determine the group index by experimentally measuring FSR.

A Michelson interferometer is composed of one splitter and two mirrors. Since the light will be reflected to the same modulated waveguide, the accumulated phase difference is twice of the MZI. Hence for the same amount of phase difference, the FSR of Michelson interferometer is half of the MZI's FSR.

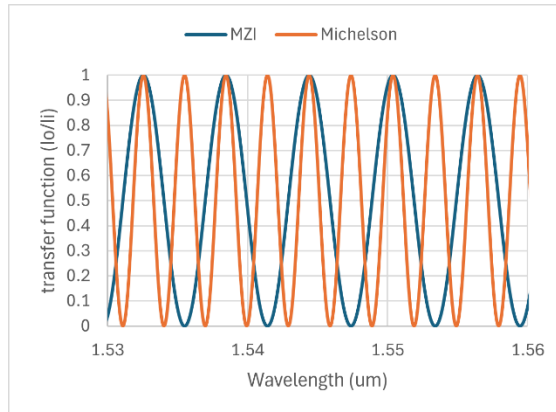

Fig. 6. Plot of the transfer function of MZI and Michelson interferometer. Assume the index at 1.5 um is 2.4, with dn/dλ = -1 μm^{-1}, $\Delta L = 100\ \mu m$.

III. Modeling and Simulation

A. Waveguide

We use Lumerical MODE solver to simulate the optical mode of the waveguide, and obtain the effective index, group index and the polynomial description of the waveguide.

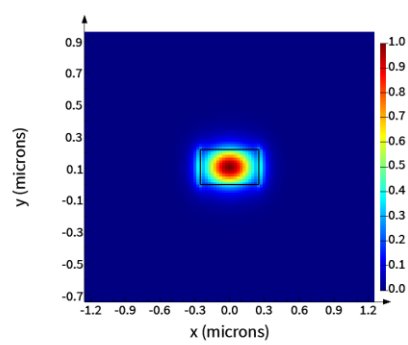

Fig. 7. TE mode profile at 1550 nm simulated using Lumerical MODE. The effective index is 2.44.

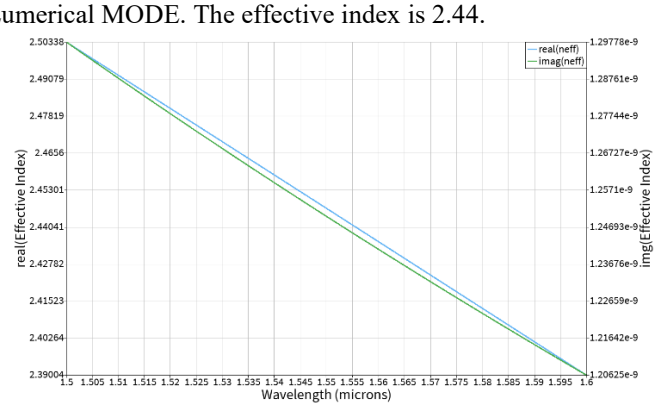

Fig. 8. Effective index dispersion of the TE mode.

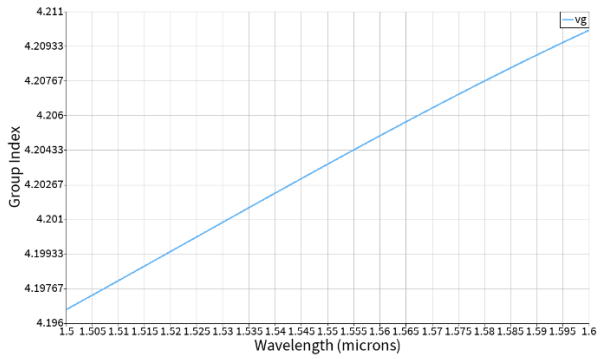

Fig. 9. Group index dispersion of the TE mode.

The polynomial model of the waveguide is

$$n_{eff}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2 \qquad (6)$$

With the coefficients

$$n_1 = 2.44682$$
$$n_2 = -1.1334$$
$$n_3 = -0.0439845$$
$$\lambda_0 = 1550\ nm$$

To characterize the manufactured waveguides, we can design a set of testing loops, with the waveguide length varying from 157 µm to 197 µm in a 20 µm increment.

B. MZI and Michelson Interferometer

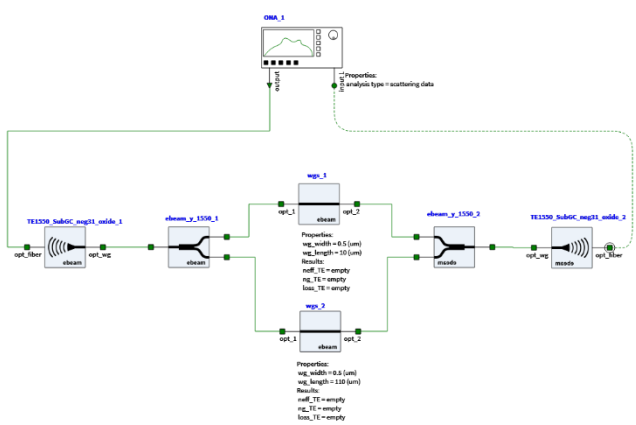

Fig. 10. The MZI model in Lumerical INTERCONNECT.

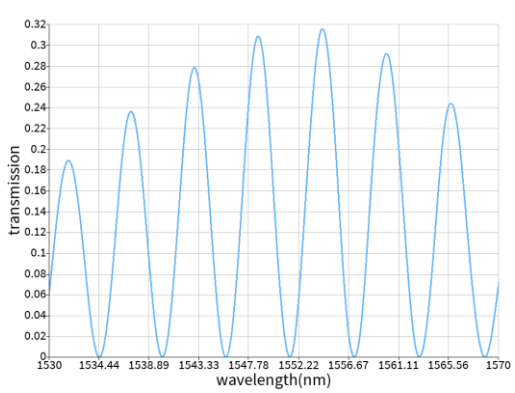

Fig. 11. The transmission spectrum of the MZI model with 100 um path difference.

TABLE I
DESIGN PARAMETERS OF MZI

ΔL (µm)	Expected FSR (nm)	Simulated FSR (nm)
0	-	-
50	19.6	11.30
100	9.8	5.71
150	6.5	3.81
200	4.9	2.86
250	3.9	2.30

From the simulated result for 50-250 µm ΔL, we can calculate the group index by either 1) curve fitting method 2) direct calculation. The calculated group indices are 4.4 and 4.2, respectively.

TABLE II
DESIGN FOR COMPARE MZI AND MICHELSON INTERFEROMETER

ΔL (µm)	MZI FSR (nm)	Michelson FSR (nm)
0	-	-
50	19.6	9.8

Use the ΔL=50 µm configuration to compare the MZI and Michelson interferometer.

IV. SUMMARY

This proposal contains 13 circuits, including 6 MZI circuits, 2 Michelson interferometer circuits, and 5 testing I/O loops. We expect to observe and calculate:

1. The transmission spectra of interferometers.
2. FSR
3. Group index
4. Comparison between the MZI and Michelson interferometer.
5. The insertion loss of I/O and propagation loss of the waveguides.

REFERENCES

[1] Lukas Chrostowski, Michael Hochberg, "Silicon Photonics Design", Cambridge University Press, 2015